Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 02)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

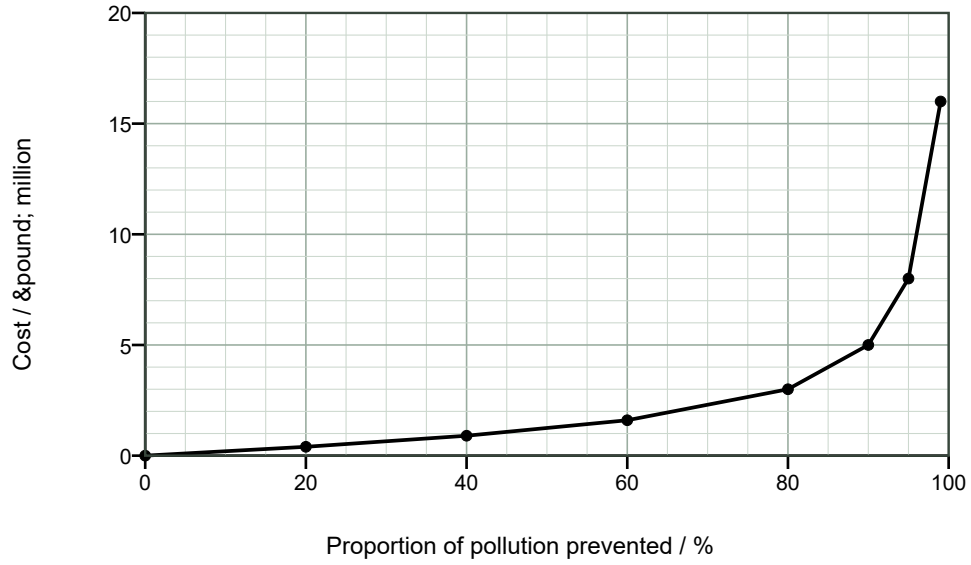
Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



0 1 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

0 1 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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0 1 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

0 1 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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0 2 . 1 Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

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Extra space

0 2 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

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0 3 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 3 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 3 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 4 A low-lying coastal town in eastern England is at risk of flooding from rising sea level and more intense rainfall.

0 4 . 1 Describe **three** ways the town could adapt to the increased flood risk.

[3 marks]

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0 4 . 2 Explain the circumstances in which managed retreat may be chosen instead of coastal defences.

[2 marks]

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0 4 . 3 Describe **two** impacts of higher temperatures on transport infrastructure.

[2 marks]

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0 4 . 4 Explain how carbon capture and storage (CCS) could reduce CO_2 emissions from a fossil fuel power station, and give **one** limitation.

[3 marks]

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0 5 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

0 5 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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0 5 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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0 5 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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0 5 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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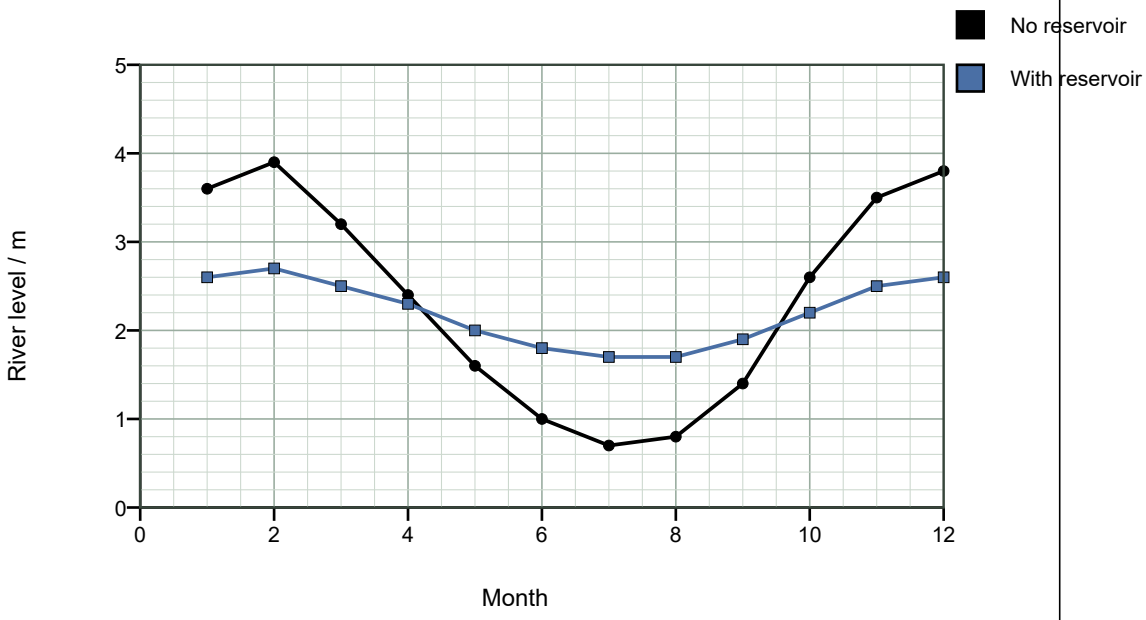
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0 6 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



0 6 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

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Answer [ ] m

0 6 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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0 6 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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0 6 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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0 7 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 7 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 7 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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0 8 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 8 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 9 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 9 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 9 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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1 0 . 1 As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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Extra space

1 0 . 2 State what is meant by the **stock** (resource base) of a mineral.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS